

Experiment 20 : Image Sharpening Using Laplacian Mask with Negative Center Coefficient

Aim

To perform **image sharpening using a Laplacian mask with a negative center coefficient** using Python and OpenCV.

Procedure

1. Read the input image.
2. Convert the image to grayscale.
3. Define a Laplacian sharpening mask with a negative center coefficient.
4. Apply the mask to the image using convolution.
5. Add the Laplacian result to the original image for sharpening.
6. Display the original and sharpened images.
7. Save the sharpened image.

Code

```
import cv2

import numpy as np

# Read input image
img = cv2.imread("input.jpg")

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Laplacian mask with negative center coefficient
mask = np.array([
    [0, -1, 0],
    [-1, 4, -1],
    [0, -1, 0]
])
```

Apply Laplacian mask

laplacian = cv2.filter2D(gray, -1, mask)

Sharpen the image

sharpened = cv2.add(gray, laplacian)

Display images

cv2.imshow("Original Image", gray)

cv2.imshow("Sharpened Image", sharpened)

Save output

cv2.imwrite("sharpened.jpg", sharpened)

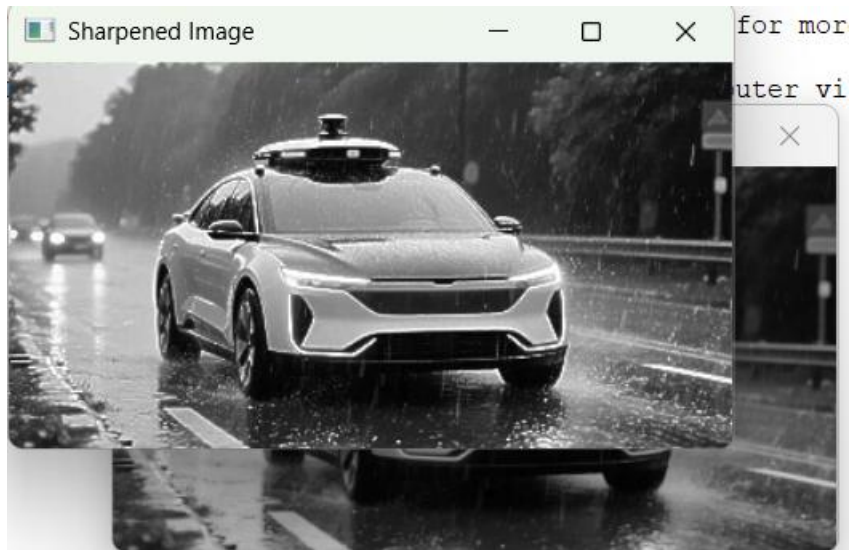
cv2.waitKey(0)

cv2.destroyAllWindows()

Input



Output



Result

Thus, image sharpening using a Laplacian mask with a negative center coefficient was successfully performed, and the edges and details of the image were enhanced.